

PCA

Give the below prompt in ChatGPT to understand Eigen value and Vector-

What is Eigen vector and Eigen value with math example?

In the context of Principal Component Analysis (PCA), the explained variance ratio refers to the proportion of the total variance in the data that is explained by each principal component. PCA is a dimensionality reduction technique that transforms a high-dimensional dataset into a lower-dimensional space, represented by its principal components.

Here's how explained variance ratio is calculated in PCA:

- 1. Compute the covariance matrix:** First, PCA calculates the covariance matrix of the original data. The covariance matrix represents the relationships and variability between different features (variables) in the dataset.
- 2. Eigenvectors and Eigenvalues:** Next, PCA finds the eigenvectors and eigenvalues of the covariance matrix. Eigenvectors represent the principal components, and eigenvalues represent the amount of variance captured by each principal component.
- 3. Normalize Eigenvalues:** The eigenvalues are typically sorted in descending order to determine the most important principal components. They are then normalized to obtain the explained variance ratio.

The explained variance ratio for each principal component is calculated as follows:

Explained Variance Ratio = Eigenvalue of the Principal Component / Sum of all Eigenvalues

The sum of all explained variance ratios across all principal components is equal to 1, as it accounts for 100% of the total variance in the original data. By examining the explained variance ratios, you can determine the importance of each principal component in capturing the variability present in the data. High explained variance ratios indicate that the corresponding principal components retain a significant amount of information from the original dataset, and they are usually the most important components in the PCA transformation. Lower explained variance ratios imply that the corresponding principal components carry less information and may not be as critical in representing the overall variability in the data.

Researchers often use the cumulative explained variance ratio to decide how many principal components to retain for dimensionality reduction. By selecting the first few principal components that cumulatively explain a large percentage of the total variance (e.g., 95% or

99%), one can achieve substantial dimensionality reduction while retaining a significant amount of the original information.

Sure, let's consider a simple example to illustrate the concept of explained variance ratio in PCA. Suppose we have a dataset with two features (variables): X and Y, and we want to reduce its dimensionality using PCA.

Here is the dataset:

Data Point	X	Y
Data Point 1	2	3
Data Point 2	1	2
Data Point 3	3	4
Data Point 4	4	5
Data Point 5	5	6

Step 1: Compute the covariance matrix

The covariance matrix for this dataset can be calculated as follows:

$$\text{Covariance Matrix} = \begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(Y, X) & \text{Var}(Y) \end{bmatrix}$$

Where:

- $\text{Var}(X)$ is the variance of feature X,
- $\text{Var}(Y)$ is the variance of feature Y,
- $\text{Cov}(X, Y)$ is the covariance between features X and Y,
- $\text{Cov}(Y, X)$ is the covariance between features Y and X.

To calculate the covariance matrix:

$$\begin{aligned}\text{Var}(X) &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \\ \text{Var}(Y) &= \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \\ \text{Cov}(X, Y) &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) \\ \text{Cov}(Y, X) &= \text{Cov}(X, Y)\end{aligned}$$

Where n is the number of data points, X_i and Y_i are the values of X and Y for the i -th data point, $\bar{X}$ and $\bar{Y}$ are the means of X and Y , respectively.

After calculating the above values, we get the covariance matrix:

$$\text{Covariance Matrix} = \begin{bmatrix} 2.5 & 3 \\ 3 & 3.5 \end{bmatrix}$$

Step 2: Eigenvectors and Eigenvalues

Next, we find the eigenvectors and eigenvalues of the covariance matrix. The eigenvalues represent the amount of variance explained by each principal component.

The eigenvalues (λ) can be found by solving the characteristic equation:

$$|\text{Covariance Matrix} - \lambda \cdot \text{Identity Matrix}| = 0$$

The eigenvalues for this example are approximately 5.677 and 0.323.

Step 3: Explained Variance Ratio

Finally, we calculate the explained variance ratio for each principal component:

The total variance in the data is the sum of all eigenvalues:

$$\text{Total Variance} = 5.677 + 0.323 = 6$$

The explained variance ratio for each principal component is:

- For the first principal component (PC1):

$$\text{Explained Variance Ratio (PC1)} = \frac{\lambda_{PC1}}{\text{Total Variance}} = \frac{5.677}{6} \approx 0.9462$$

- For the second principal component (PC2):

$$\text{Explained Variance Ratio (PC2)} = \frac{\lambda_{PC2}}{\text{Total Variance}} = \frac{0.323}{6} \approx 0.0538$$

So, in this example, the first principal component (PC1) explains approximately 94.62% of the total variance in the data, while the second principal component (PC2) explains approximately 5.38% of the total variance.

The takeaway from the above example about the explained variance ratio in PCA is as follows:

1. Interpreting the Explained Variance Ratio: The explained variance ratio tells us the proportion of total variance in the data that is captured by each principal component. In the example, PC1 explains about 94.62% of the total variance, while PC2 explains about 5.38% of the total variance. This indicates that the first principal component (PC1) contains a significant amount of information about the original dataset, while the second principal component (PC2) captures a smaller amount of information.

2. Dimensionality Reduction: PCA is often used for dimensionality reduction. In this example, since PC1 explains the majority of the variance (94.62%), we could potentially retain only PC1 and discard PC2 without losing much information. This reduces the dimensionality of the dataset from two dimensions (X and Y) to just one dimension (PC1). This can be particularly helpful when dealing with high-dimensional datasets, as it allows us to represent the data in a lower-dimensional space while preserving most of its variability.

3. Choosing the Number of Principal Components: The explained variance ratio is used to determine how many principal components to retain. A common approach is to choose the number of principal components that cumulatively explain a desired percentage of the total variance. For instance, if you want to retain 95% of the total variance, you would sum the explained variance ratios starting from the first principal component until you reach or exceed 95%.

4. Visualization : PCA is often used for visualization purposes as well. Since principal components are orthogonal (uncorrelated) to each other, the first few principal components can be used to visualize the data in a reduced-dimensional space. In this example, we could plot the data points along the first principal component (PC1) axis, providing a simplified representation of the original dataset.

Overall, the explained variance ratio in PCA provides valuable insights into the importance of each principal component and aids in making decisions regarding dimensionality reduction

and data visualization. It allows us to understand how much information is retained in each principal component and guides us in selecting the appropriate number of principal components to achieve the desired level of dimensionality reduction while preserving the key patterns in the data.